



IndianOil

Indian Oil Corporation Limited
Northern Region Pipelines, Rewari

Ref: NRPL/REW/MECH/2018/3013/02

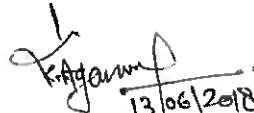
Date: - 13.06.2018

Sub: Major Maintenance Report of SMPL MP1 NRPL, Rewari Station

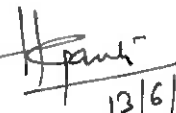
The major maintenance of MP1 SMPL was carried out from 06.06.2018 to 09.06.2018 due to the high vibration at pump DE Horizontal.

The pump trial was taken on 09.06.2018 and checked vibration with all other performance parameters of pump & found satisfactory.

The detailed report of pump is enclosed herewith for review please.


13/06/2018
(Yogesh Agarwal)
O&ME, Rewari

CMNM, Rewari


13/6/18

GM, Rewari


13/6/18

CGM (O)

SUB: OVERHAULED MAINTENANCE REPORT OF SMPL MP1 REWARI STATION

The detail specification of the Mainline pump SMPL MP1 Rewari Station is as follows:

MAINLINE PUMP SMPL MP1			
MAKE :	BPCL	TYPE	DVDS 14X14X17.5, Single stage
NO. OF UNITS :	03 (MP1, MP2 & MP3)	S. NOS.	202012006-04/01 (P.O NO: 23573216 Dated; 19.03.2012)
CAPACITY :	1037 M3/Hr. 1st Stage, 1151 M3/Hr. IInd Stage	HEAD	215 M 1st Stage 251 M IInd Stage
RPM :	Max. 2770	SP. GR.	Product Crude oil, SP. GRAVITY 0.88
DE BEARING :	Split Bush bearing	NDE BEARING	Split Bush bearing +Tilting Pad thrust bearing
IMPELLER DIA :	460 MM	ROTATION	CCW
EFFICIENCY:	65%/82% (1st & IInd stage)	RECOMMEND LUBE OIL	SERVO SYSTEM 46, Forced lubrication
DRAWING NO	2CP 11891 Cross Section	Date of Commissionin	Year 2015
MECH.SEAL:	Type: UOP, 4.250" Single Inside Cart. Drg. No. 2H-133298/99 R2 BPCL Ref. Drg.4CP-60287/88 REV-1	COUPLING:	EMH-390(460) I/I/5078 CPLG Rathi Transpower Makes Drawing No: R/LM-5078/2 ,DBSE 460 mm
PUMP SUCTION/ DISCHARGE SIZE	14"X600#	VALVE SIZE	18"X600# CCFV Makes
Rotor Weight	167 kg, Bal. spec. ISO 1940 Gr.2.5	Motor details	Squirrel cage induction Motor, Type: DHSL-560LK-02A, 995 KW, 6.6 KV, Max speed-2988 rpm, Full load current 98 A, eff%-96.8%, Makes: LOHER, Germany

Introduction: Three SMPL Mainline Pumping Units are installed at NRPL Rewari station for the boosting of crude oil into Rewari –Panipat section. The SMPL Mainline Pumping units were commissioned in Oct 2015.

Running hours of MP1 at the time of Maintenance: 6754.00 hrs.

Reason of Maintenance/ Problem: The pump was showing high vibration (i.e. 5.1 to 6.4 mm/sec rms) at pump DE horizontal above 2200 rpm. First doubt gone upon suction side conical strainer as due to low flow in suction may be reason of high vibration. First suction side installed conical strainer was opened to check the condition of filter screen. Upon opening it was noticed that conical strainer got damaged & front side perforated sheet portion missing.

Hence, it was decided to check the condition of internal parts by lifting the top casing of pump.

Maintenance activity: During the pump maintenance the following activities were carried out.

On 06.06.2018

1. Work permit taken from operation department and electrically isolation was done for Pump Motor, L.O Motor, Suction and discharge valves of the pump.
2. Both side valves checked for passing found OK.
3. Crude oil drained out and Pump de-coupled.
4. Both side bearing housing & seals were removed.

On 07.06.2018

5. Pump casing stud removed and top cover of pump lifted.
6. Pump rotor assembly checked and conical strainer perforated pieces were found in stuck into the suction eye of impeller DE side.
7. Complete assembly lifted safely from position to remove the perforated piece.
8. Conical strainer perforated screen pieces were removed safely and cleaning of impeller and other rotor parts was carried out.
9. Pump lower parts volute checked and cleaned.
10. Pump rotor assembly placed on position and top cover fitted.

[Signature]

On 08.06.2018

11. All the casing studs were tightened.
12. Both side seal were dismantled, all the internal parts were cleaned & defective parts of DE side seal such as Rotary seal ring, stationary seal ring and 'O' ring set were replaced.
13. Both side seals were fitted on pump, all the connection of seal cooling line completed.
14. Pump charged with crude oil to check the any kind of leakage with pressure. Pump checked at 20.0 kg/cm² pressure and No-leakage was noticed.
15. Both side bearing housing were fitted.

On 09.06.2018

16. All the lube oil circulation and drain line connection were fitted.
17. Lube oil pump run for ½ hour to set the lube oil pressure and flow of lube oil in bearing housing.
18. Pump alignment was done and coupling fitted.
19. Pump trial was taken on load and pump vibration found normal at high rpm. (i.e. 1.1 to 1.7 mm/sec rms at 2200 rpm to 2770 rpm)
20. Pump handed over to operation.

During dismantling of the pump following defects were observed.

1. Conical strainer perforated screen piece were found stuck into suction eye of impeller at Pump DE side.
2. DE side seal parts such Rotary seal ring, stationary seal ring and 'O' ring set were found defective the same were replaced with new parts (seal spares received with pump used).

Estimated cost of Spare parts used during the maintenance of pump

SNO	SAP CODE	ITEM DISCRIPTION	QTY	REMARKS	Approx. cost of parts landed cost taken.
1.	1726840114	Rotary seal ring Si-Carbide	01	Seal spares received with pump	AS per PO of pump
2.	1726840114	Stationary seal ring Carbon	01	Seal spares received with pump	Approx. Value of parts
3.	1726840124	'O' ring set	01	Seal spares received with pump	RS: 165000.00

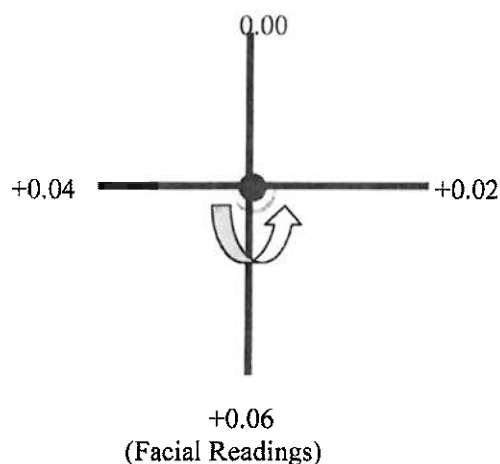
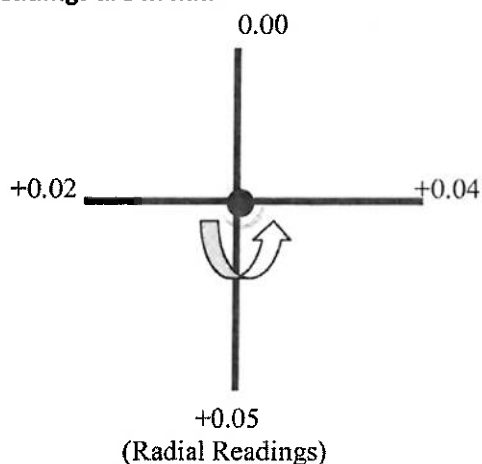
Cost of Maintenance: RS 1,65,000.00 (One lac Sixty Five Thousand Only)

Clearance report is enclosed as annexure-I

ALIGNMENT READINGS AFTER MAINTENANCE TAKEN ON 06.11.2015

Location of readings	0 degree	90 degree	180 degree	270 degree
Radial readings	0.00	+0.04	+0.05	+0.02
Facial readings(Axial readings)	0.00	+0.02	+0.06	+0.04

*All the readings are in mm



K. Aggarwal

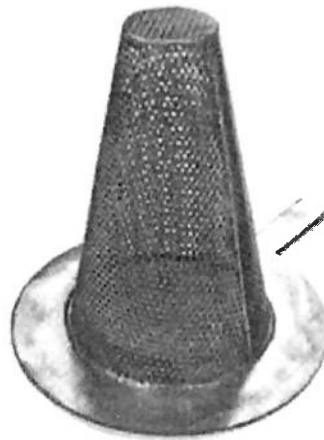
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VIBRATION READINGS DETAILS OF PUMP

RPM	BEFORE MAINTENANCE						AFTER MAINTENANCE CHECKED ON 09.06.2018					
	PUMP Vibrations in rms mm/sec Velocity						MOTOR Vibrations in rms mm/sec Velocity					
	DE			NDE			DE			NDE		
	H	V	A	H	V	A	H	V	A	H	V	A
2000	1.2	0.8	0.4	1.3	1.0	0.3	1.1	0.6	0.5	1.1	0.4	0.6
2200	4.5	1.1	1.3	3.5	1.4	1.5	1.1	0.7	1.0	1.2	0.6	0.9
2400	5.1	1.4	1.3	4.6	1.7	1.4	1.2	0.8	0.7	1.1	0.7	0.7
2500	5.2	1.6	1.3	4.7	2.4	1.3	1.5	0.9	0.7	1.6	0.7	0.5
2600	5.3	2.1	1.6	4.9	2.5	1.8	1.3	1.0	0.9	1.5	1.6	0.7
2700	5.6	2.1	1.6	5.0	2.5	1.8	1.4	1.1	0.9	1.6	1.5	1.0
2770	6.4	2.3	1.8	5.0	3.4	1.9	1.6	1.3	1.3	1.7	1.5	1.1

After the overhauled maintenance of pump the vibration readings of pump were reduced and came between the Normal limits (0.0 to 4.5 mm/sec rms value) checked as per the ISO -10816 class-III (revised old ISO 2372) vibration severity standard of rotary equipment.

Root cause analysis of the failure of strainer & pump: The conical strainer was fitted during the commissioning of pump fabricated with perforated sheet 6.0 mm hole dia and layer with 40 wire mesh. Due to sudden chocking of 40 wires mesh the conical strainer was ruptured and front nose pieces moved forward and stuck into impeller suction eye. Due to these pieces there was rotor unbalance and hence developing high vibration above 2200 rpm.

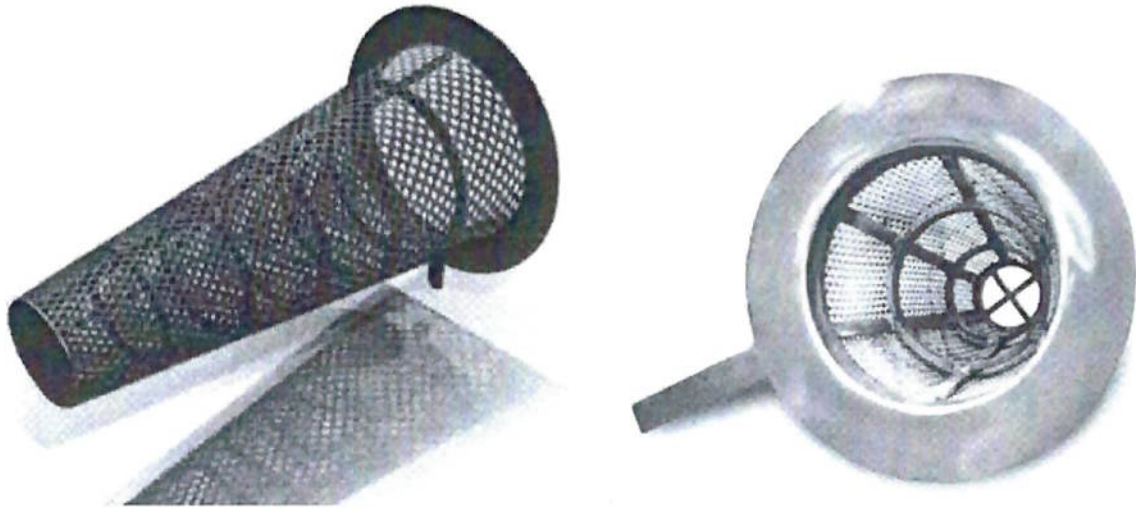


Design of conical strainer

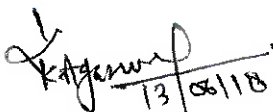
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Future remedial Action plan:

1. The previously fitted temporary conical strainer was without internal frame support. Hence, we are fabricating new conical strainer with internal frame support for strengthening.
2. New temporary Conical Strainer will be fabricated with perforated sheet (1/4" hole dia x 5/16" pitch x 1/8" thickness), No wire mesh will be provided on perforated screen as pump already run 6754 hrs. , SMPL-DBL commissioning completed since long back & very small quantity of muck receiving during pigging.
3. As per the approved guidelines of PLHO (Ref PL/PJ/M/1.0 dated: 31.07.2017) regarding the removal of temporary conical strainer post commissioning of mainline pumping units. According to the guidelines conical strainer can be removed from suction arm of pump with 720 hrs. Operation without chocking since last cleaning. SMPL MP1 running hrs. is 6754.00
4. We have planned to open SMPL MP2 & MP3 conical strainer for cleaning & inspection to avoid this kind of problem in future.



New Design of conical strainer


(Yogesh Agarwal)
O & ME, NRPL Rewari

Enclosed:

- Clearance report
- Defective parts and defects details
- Internal parts details